

Ibrahim Hussain

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PROFILE

Data Science undergraduate at FAST University, Islamabad (graduating 2027), with industry experience in multi-agent AI systems, RAG pipeline engineering, large-scale financial data modelling, and end-to-end analytical system design. Built and shipped AI systems beyond academic settings — including a live multi-agent health assistant at SmartForum and a multimodal deep learning framework for financial risk management integrating 25 years of U.S. market data. Combines statistical rigour with practical engineering across finance, healthcare, and decision analytics domains.

EXPERIENCE

AI Engineer Intern — SmartForum

June – September 2025

Islamabad, Pakistan · Hybrid

- Engineered and optimised RAG pipelines in a production environment using PostgreSQL indexing and query optimisation, reducing retrieval latency by ~20% and improving response accuracy for real user queries.
- Built modular multi-agent AI workflows using LangChain and LangGraph, coordinating agent-to-agent communication, task routing, and data-processing automation across a live multi-agent health assistant system.
- Integrated Automatic Speech Recognition (ASR) capabilities into the agentic pipeline, enabling real-time voice interaction in a production deployment.
- Designed relational database schemas and Flask-based backend workflows to support agent coordination, state management, and data persistence.

PROJECTS & TECHNICAL EXPERIENCE

Explainable Financial Risk Management Framework

January – May 2026

AI Systems & Data Engineering Lead

- Designed a multimodal deep learning architecture combining a temporal attention encoder, FinBERT text encoder for SEC filing analysis, and Graph Neural Network modules (StemGNN, MTGNN) adapted for contagion detection and market-regime identification.
- Engineered large-scale financial data pipelines across 25 years of U.S. market data (2,500 stocks × 6,286 trading days), processing 17GB+ of SEC EDGAR filings and 900k+ structured submission records with zero missing values in the final panel.
- Built leakage-safe chronological training splits across 2000–2024 and produced comprehensive technical documentation covering architecture decisions, XAI design, and evaluation methodology.

Writer's Block — Multi-Agent LLM Pipeline

December 2025

AI Systems Engineering

- Architected a production-grade six-agent LLM system for automated content generation, implementing a MapReduce-style orchestration pattern with parallel agent execution, auto-fallback model routing, and retry logic — achieving an 85.5% task success rate.
- Engineered cost-optimised model routing across multiple LLM providers, reducing per-article generation cost to ~\$0.04 through intelligent task-to-model assignment.

- Built full production stack: Flask backend, Supabase/PostgreSQL database layer for real-time job tracking, compileragent for multi-agent output fusion, and guardrails for quality enforcement.

Inflation Dynamics in Pakistan — Macroeconomic Forecasting

May 2025

Statistical Modelling & Time Series Analysis

- Collected, cleaned, and validated 43 years of Pakistani macroeconomic data across multiple sources, handling missing values through structured imputation and cross-source reconciliation.
- Applied ARIMA, Ridge, Lasso, and Elastic Net regression models with AIC, RMSE, and MSE-based model selection; identified exchange rate volatility and money supply growth as primary inflation drivers.

Disaster Recovery & Resilience Analysis (RRS)

December 2025

Data Engineering & Predictive Modelling

- Integrated 7,000+ observations across four independent data sources (disaster records, GDP, governance indices, HDI), designed a Resilience Recovery Score by modelling pre- and post-disaster GDP dynamics, and built interactive visual analytics to communicate findings.

TCGA Multi-Omics Integrative Analysis

November 2025

Data Engineering & Machine Learning

- Processed and integrated multi-omics cancer datasets (genomics, transcriptomics, epigenomics) with 20GB+ of histopathology images, building high-dimensional feature pipelines and classification models for cancer stage and risk prediction.

EDUCATION

FAST University (NUCES)

August 2023 – Present

Bachelor of Science in Data Science — Islamabad, Pakistan (Expected Graduation: 2027)

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Big Data Systems, Database Systems & Warehousing, Advanced Statistics, Operating Systems, Artificial Intelligence, Parallel & Distributed Computing, Software Engineering.
- Awards: Winner — Code Jail Competition, TechConnect by Fast Computing Society in November 2024.

The Science School

May 2021 – August 2023

GCE A-Level — Islamabad, Pakistan

Subject	Grade
Mathematics	B(b)
Computer Science	C(c)
Physics	B(b)

The Science School June 2020 – June 2021

Cambridge O-Level & IGCSE — Islamabad, Pakistan

Subject	Qualification	Grade
Mathematics	O-Level	A*(a*)
Computer Science	O-Level	A(a)
English Language	O-Level	B(b)
Physics	IGCSE	A*(a*)

Chemistry	IGCSE	A(a)
Islamiyat	O-Level	B(b)
Pakistan Studies	O-Level	B(b)
Second Language Urdu	O-Level	B(b)

VOLUNTEERING

Volunteer Educator — SAYA School (SAYA Welfare Society)

Islamabad, Pakistan · Education & Community Outreach

January – May 2025

- Volunteered as part of a six-member team at SAYA School, a welfare initiative providing free education to underprivileged children on the outskirts of Islamabad, founded by Dr. Tughral Yamin under the SAYA Welfare Society.
- Personally taught Mathematics and English to Class 5-A across multiple sessions, adapting delivery in real time to varying learning levels and adapting pedagogical approach to the specific needs of primary school students.
- Co-delivered weekly sessions across multiple grade levels covering academic revision, career counselling, climate change awareness, freelancing as a digital career path, basic electronics and chemistry, STEM exploration, and recycling/sustainability activities.
- Student impact included spontaneous climate action taken at home, shared during school morning assemblies, and strong engagement with career and digital literacy content from students with minimal prior exposure to these topics.

TECHNICAL SKILLS

Skill	Experience
AI & LLM Systems	LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipeline Design, Prompt Engineering, LLM Cost Optimisation, Agentic Workflow Design
Machine Learning & DL	Transformers, BERT/FinBERT, Graph Neural Networks (StemGNN, MTGNN), LSTMs, Supervised Learning, Fine-tuning, NLP, Classification
Data Science	Statistical Modelling, Regression Analysis (Ridge, Lasso, Elastic Net), Hypothesis Testing, Time Series Forecasting, ARIMA, Feature Engineering, Data Cleaning
Data Engineering	PostgreSQL, MySQL, Large-Scale Pipelines, Multi-Source Data Fusion, Vector Databases (Pinecone), Time-Series Engineering, Relational Schema Design
Programming & Tools	Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL, R, C++, Flask, Git/GitHub, REST APIs, Linux, Jupyter Notebooks
Speech & Real-Time AI	Automatic Speech Recognition (ASR) integration, real-time pipeline design

CERTIFICATIONS

Award / Certificate	Issuing Organisation	Date
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Winner — Code Jail Competition	FAST Computing Society	November 2024
Intro to Python Workshop	Protect Lab (Summer Workshop)	July 2024
Astronomy Camp	NCGSA / Institute of Space Technology	March 2022
Space Summer School	Institute of Space Technology, Islamabad	July 2017
2nd Position — STEM Project Competition	The Science School	December 2019

LANGUAGES

English <i>Fluent (Professional)</i>	Urdu <i>Native</i>	Italian <i>Elementary</i>
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